

8th Grade Unit 6 Assessment Guide

General Information	The assessment covers the benchmarks MA.8.AR.3.1, MA.8.AR.3.2, and MA.8.GR.2.4. These benchmarks are important foundations for unit 7. In unit 7, students determine slope from tables, graphs, or as a written description. In unit 6, students relate slope to the proportional relationship of the vertical distance and the horizontal distance of similar triangles that are drawn on the coordinate grid. Students also bridge their learning from grade 7 of proportional relationships to build a foundation for writing linear equations. A student who knows how to find slope (MA.8.AR.3.2) and understands it as the ratio of $\frac{\text{vertical distance}}{\text{horizontal distance}}$ will not have to depend on the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Teachers may want to consider how students who struggle on the questions for MA.8.AR.3.1 may benefit from reviewing content from unit 11 of grade 7. Small group differentiation before unit 7 or simultaneously with lessons in unit 7 may help students bridge any gaps in their learning. Students who struggle with questions for MA.8.GR.2.4 may benefit from reviewing content from unit 5 of grade 7.
Calculator Information	The questions on this assessment are calculator neutral so students may benefit from having access to scientific calculators for this assessment, but a student does not need a calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1a	Benchmark(s)	MA.8.AR.3.2
	The ratios ☐ $\frac{DE}{AC}$ ☐ $\frac{AB}{CB}$ ☒ $\frac{AC}{AB}$ ☐ $\frac{CD}{CE}$ and ☒ $\frac{CD}{DE}$ ☐ $\frac{DE}{CE}$ ☐ $\frac{AB}{EB}$ ☐ $\frac{CE}{CB}$ are equivalent to the slope of the line CB.	Recommended Scoring (12 Total Points)	Recognizing the slope triangle is an important concept from this unit. Each correct selection is 6 points.
Question	1b	Benchmark(s)	MA.8.AR.3.2
	$-\frac{4}{10}$	Recommended Scoring (8 Total Points)	Accept equivalent values for the slope.

Question	2	Benchmark(s)	MA.8.GR.2.4
$\frac{50}{150} \text{ or } \frac{54}{162}$		Recommended Scoring (6 Total Points)	Accept equivalent values.
Question	3	Benchmark(s)	MA.8.GR.2.4
12		Recommended Scoring (10 Total Points)	Consider giving students 4 points if they correctly set up a proportion. Monitor student work for minor errors that the student carries through the remainder of their work. Consider deducting no more than 2 points for such minor errors.
		Additional Notes	Accept equivalent values.

Question	4a		Benchmark(s)	MA.8.AR.3.1								
<table><tr><td rowspan="2">Independent Variable</td><td rowspan="2">number of balloons</td><td>Letter</td></tr><tr><td>Answers will vary.</td></tr><tr><td rowspan="2">Dependent Variable</td><td rowspan="2">weight in pounds</td><td>Letter</td></tr><tr><td>Answers will vary.</td></tr></table>			Independent Variable	number of balloons	Letter	Answers will vary.	Dependent Variable	weight in pounds	Letter	Answers will vary.	Recommended Scoring (12 Total Points)	Students may simply write “balloons” and “weight”. Teachers can determine if they wish to assign points to providing a letter in this part or they wish to just monitor that students use the appropriate letter in question 4b.
Independent Variable	number of balloons	Letter										
		Answers will vary.										
Dependent Variable	weight in pounds	Letter										
		Answers will vary.										
Question	4b		Benchmark(s)	MA.8.AR.3.1								
$y = \frac{51}{54}x$			Recommended Scoring (8 Total Points)	Accept equivalent equations that match the letters students decided to use. If students are given calculations, determine what equivalent values for the slope are accepted.								

Question	5a	Benchmark(s)	MA.8.AR.3.1
The relationship between time and the height of the bamboo proportional because for all of Allyson's measurements.	☐ is ☐ is not ☐ time ☐ the ratio $\frac{y}{x}$ ☐ the bamboo height ☐ increases by the same amount ☐ is not the same	Recommended Scoring (12 Total Points)	Consider giving 6 points for the first selection "is not" and 6 points for the correct combination for second and third boxes.
Question	5b	Benchmark(s)	MA.8.AR.3.1
Answers will vary.		Recommended Scoring (6 Total Points)	Possible work could be that the student shows the ratio $\frac{\text{bamboo height}}{\text{time}}$ is not the same for each set of values.
Question	6	Benchmark(s)	MA.8.GR.2.4
46		Recommended Scoring (10 Total Points)	Consider giving students 4 points if they correctly set up a proportion. Monitor student work for minor errors that the student carries through the remainder of their work. Consider deducting no more than 2 points for such minor errors.

Question	7	Benchmark(s)	MA.8.AR.3.1
proportional		Recommended Scoring (4 Total Points)	Partial credit is not recommended.
Question	8	Benchmark(s)	MA.8.AR.3.1
not proportional		Recommended Scoring (4 Total Points)	Partial credit is not recommended.
Question	9	Benchmark(s)	MA.8.AR.3.1
not proportional		Recommended Scoring (4 Total Points)	Partial credit is not recommended.
Question	10	Benchmark(s)	MA.8.AR.3.1
proportional		Recommended Scoring (4 Total Points)	Partial credit is not recommended.